Harshith Kurakula

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SUMMARY

Software Engineer with 3+ years of experience in building automation tools, managing distributed systems, optimizing database performance, and improving application reliability in cloud environments. Proven expertise in streamlining workflows, enhancing system performance and scaling systems effectively. Skilled in Python, Java, SQL, AWS, CI/CD pipelines and Docker. Actively seeking summer internships in 2026.

EDUCATION

Arizona State University — *Tempe, AZ, USA*
Master of Science in Computer Science, **Current GPA - 4.00 / 4.00** — **August 2025 – December 2026**

Relevant Coursework: Data Mining, Statistical Machine Learning, Data Visualization, Semantic Web Mining, Knowledge Representation

Vellore Institute of Technology — *Vellore, Tamil Nadu, India*
Bachelor of Technology in Computer Science and Engineering, **GPA - 3.84 / 4.00** — **July 2018 – June 2022**

Relevant Coursework: Data Structures, Algorithms, Database Systems, Object-Oriented Programming, Web Development

SKILLS

Programming Languages:	Python, Java, SQL, C, C++, JavaScript, TypeScript, R
Cloud, DevOps & Databases:	AWS, GCP, Git, Docker, Kubernetes, Terraform, CI/CD, Splunk, MySQL, PostgreSQL, MongoDB
Web Development:	HTML5, CSS3, Flask, Django, REST APIs, Vue.js, Node.js, React, Next.js, Tailwind CSS
ML & Data Tools:	NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Tableau, Grafana

EXPERIENCE

Chargebee Technologies — *Chennai, Tamil Nadu, India*

Senior Software Engineer (SRE Team) — **April 2025 – July 2025**

- **Optimized** database connection management by consolidating schema-level connection pools into a shared pool per **AWS RDS** instance, **reducing connection spikes and idle connections by 20%** and improving resource utilization, scalability, and system stability.
- **Automated** configuration onboarding through a centralized database configuration management system, enabling consistent configuration across microservices and reducing **manual effort by 40% during infrastructure provisioning**.

Software Engineer (SRE Team) — **August 2022 – March 2025**

- **Led** AWS RDS database upgrade initiatives using Blue-Green deployment strategies, coordinating planning, execution, and validation to achieve **near-zero downtime**, preserve data integrity, and **reduce service disruption by 90%** across critical production systems.
- **Built** an automated schema comparison tool using Java and SQL to detect database inconsistencies across environments, capturing all schema mismatches while **reducing manual debugging time by 30%** and improving overall production reliability.
- **Implemented** an API feature for an internal database migration tool to automate schema changes following RDS provisioning and sharding activities. This streamlined setup **reduced manual intervention by 25%** and improved system efficiency and reliability.
- **Collaborated** with cross-functional engineering, platform, and product teams to embed SRE best practices, **reducing incident recurrence by 30%**, improving system stability, and enabling scalable multi-region deployments.

Software Engineer Intern (SRE Team) — **January 2022 – July 2022**

- **Engineered** a centralized observability and monitoring dashboard integrating Splunk, ELK Stack, and database logs, enabling proactive incident detection and **reducing Mean Time to Resolve (MTTR) by 25% while maintaining 99.9% system availability**.
- **Spearheaded** infrastructure provisioning automation using AWS RDS, DynamoDB, and CloudFormation, **reducing manual configuration effort by 40%**, **accelerating deployment time by 30%**, and improving cross-environment reliability and consistency.

PROJECTS

Multimodal Image Retrieval System *(Python, Keras, Flask, Vue.js)*

- **Designed and implemented** a multimodal deep learning retrieval pipeline leveraging ResNet (CNN) for visual feature extraction and LSTM-based text embeddings, achieving **80–85% retrieval accuracy** by effectively aligning visual and semantic representations.
- **Built** an end-to-end image–text retrieval workflow integrating advanced text preprocessing and image feature analysis, significantly **outperforming traditional baseline models** in search relevance and precision.

Facial Emotion Recognition *(Python, TensorFlow, Keras, OpenCV)*

- **Engineered** a real-time facial emotion recognition system using OpenCV and Deep Convolutional Neural Networks (DCNNs) to classify **7 facial expressions**, optimizing the model to capture rapid facial dynamics with **higher accuracy than traditional baselines**.
- **Developed** a real-time alerting and data transmission module to stream emotion predictions and geolocation metadata, enabling immediate actionable insights for remote monitoring and safety applications.